

Laboratorio: Selección y Consulta de Datos (DQL)

Atributo	Detalle
Dificultad	Introductorio
Tiempo Estimado	45 minutos
Servicios Principales	Amazon EC2 (Command Host), MySQL Engine, Data Query Language (DQL)

Resumen y Objetivos

Este laboratorio se centra en el **Lenguaje de Consulta de Datos (DQL)**. Utilizarás la sentencia **SELECT** como herramienta principal para extraer información específica de una base de datos relacional. Aprenderás a filtrar, ordenar y agregar datos para transformar registros crudos en información útil para la toma de decisiones.

Al finalizar este laboratorio, serás capaz de:

- 1. Emplear la sentencia **SELECT** para recuperar datos de tablas específicas.
- 2. Utilizar funciones de agregación como **COUNT()** para obtener métricas rápidas.
- 3. Aplicar operadores lógicos (**AND**) y de comparación (**>**, **<**, **=**) para filtrar resultados.
- 4. Implementar cláusulas de ordenamiento (**ORDER BY**) y alias de columna (**AS**) para mejorar la legibilidad de los reportes.

Análisis del Escenario

El equipo de operaciones ya ha poblado la base de datos **world** con información global sobre países y ciudades. Tu rol como analista de datos es validar esta información y responder a solicitudes de negocio específicas mediante consultas SQL. El diagnóstico inicial indica que los datos están estructurados, pero requieren filtrado avanzado para responder preguntas complejas sobre demografía y geografía regional.

Arquitectura

La infraestructura se mantiene bajo el modelo de administración segura:

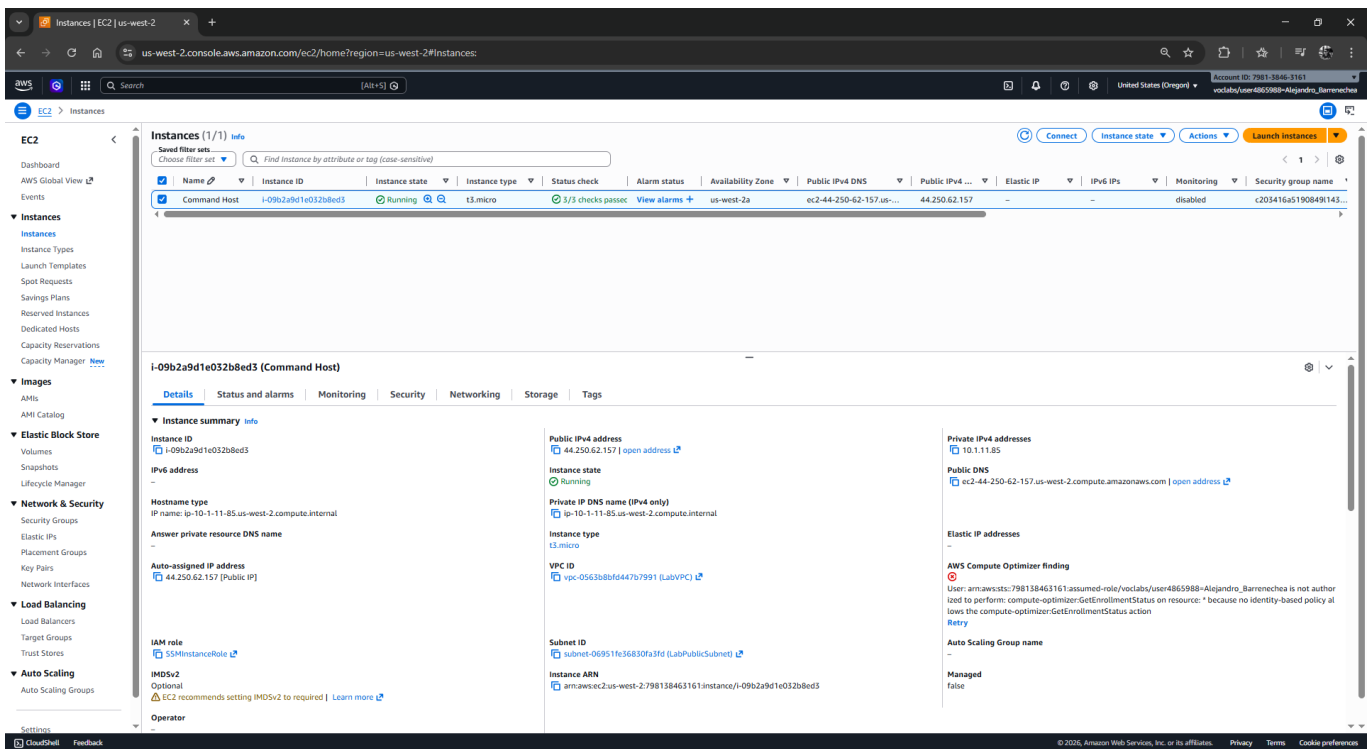
- **Command Host:** Instancia EC2 que sirve como punto de entrada para el administrador.
- **Motor MySQL:** Procesa las consultas enviadas por el usuario y genera el *Result Set* (conjunto de resultados).
- **Base de Datos world:** Contiene las tablas **country**, **city** y **countrylanguage**.

Desarrollo de las Tareas Paso a Paso

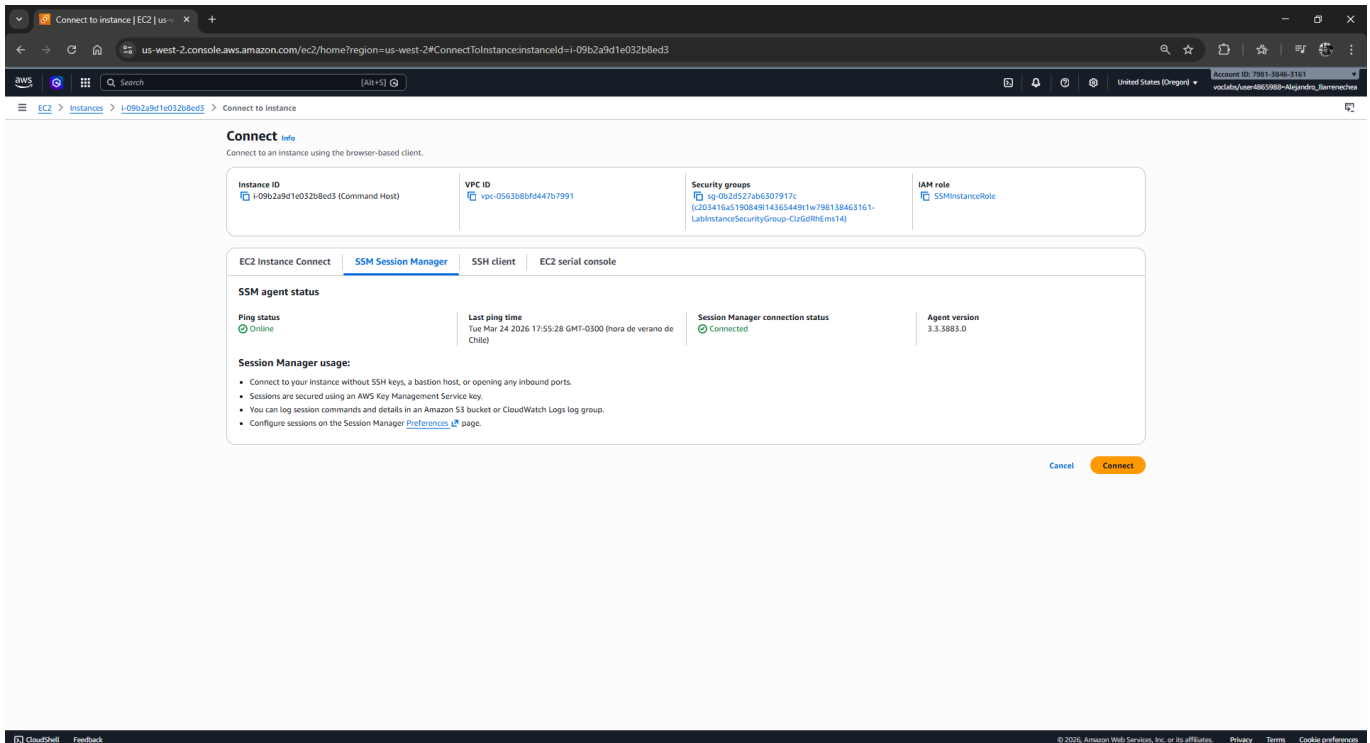
Tarea 1: Conexión al Command Host

Establece el entorno de trabajo seguro para interactuar con el motor de base de datos.

1. Navega a la consola de **Amazon EC2**.
2. En el panel izquierdo, selecciona **Instances**.
3. Selecciona la instancia **Command Host**, haz clic en el botón **Connect** (Conectar).



4. Elige la pestaña **Session Manager** y presiona **Connect**.



5. En la terminal, prepara el entorno con los siguientes comandos:

```
sudo su
cd /home/ec2-user/
```

6. Accede al shell de MySQL con las credenciales de laboratorio:

```
mysql -u root --password='re:St@rt!9'
```

Instances | EC2 | us-west-2

Systems Manager | us-west-2

us-west-2.console.aws.amazon.com/systems-manager/session-manager/i-09b2a9d1e032b8ed3?region=us-west-2#

Session ID: user4865988=Alejandro_Barrenechea-sgqekfpqzrscens95ir3esxfu

Shortcuts

Instance ID: i-09b2a9d1e032b8ed3

```
sh-4.2$ sudo su
[root@ip-10-1-11-85 bin]# cd /home/ec2-user/
[root@ip-10-1-11-85 ec2-user]# mysql -u root --password='re:St@rt!9'
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 5
Server version: 10.5.29-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]>
```

Tarea 2: Consulta de la Base de Datos world

Realizarás extracciones de datos incrementando la complejidad de los filtros.

1. Confirma la disponibilidad de la base de datos:

```
SHOW DATABASES;
```

Instances | EC2 | us-west-2

Systems Manager | us-west-2

us-west-2.console.aws.amazon.com/systems-manager/session-manager/i-09b2a9d1e032b8ed3?region=us-west-2#

Session ID: user4865988=Alejandro_Barrenechea-sgqekfpqzrscens95ir3esxfu

Shortcuts

Instance ID: i-09b2a9d1e032b8ed3

Terminate

```
1522 | TF | Oceania | Polynesia | 650.00 | 1970 | 99000 | 67.9 | 146.00 | 170.00 | Tonga | Monarchy
3334 | TO | North America | Caribbean | 5130.00 | 1962 | 1295000 | 68.0 | 6232.00 | 5867.00 | Trinidad and Tobago | Republic
3336 | TT | Africa | Northern Africa | 163610.00 | 1956 | 9586000 | 73.7 | 20026.00 | 18898.00 | Tunis/Tunisie | Republic
3349 | TR | Asia | Middle East | 774815.00 | 1923 | 66591000 | 71.0 | 210721.00 | 159122.00 | Türkiye | Republic
3358 | TR | Oceania | Polynesia | 26.00 | 1978 | 12000 | 66.3 | 6.00 | NULL | Tuvalu | Constitutional Monarchy
3424 | TV | Asia | Eastern Asia | 36188.00 | 1945 | 22256000 | 76.4 | 256254.00 | 263451.00 | T'ai-wan | Republic
3263 | TZ | Africa | Eastern Africa | 883749.00 | 1961 | 33517000 | 52.3 | 8905.00 | 7388.00 | Tanzania | Republic
3306 | TZ | Africa | Eastern Africa | 241038.00 | 1962 | 21778000 | 42.9 | 6313.00 | 6887.00 | Uganda | Republic
3425 | UG | Europe | Eastern Europe | 603700.00 | 1991 | 50456000 | 66.0 | 42168.00 | 49677.00 | Ukraine | Republic
3426 | UA | Oceania | Micronesia/Caribbean | 16.00 | NULL | 0 | NULL | 0.00 | NULL | United States Minor Outlying Islands | Dependent Territory of the US
NULL | UM | South America | South America | 175016.00 | 1828 | 3337000 | 75.2 | 20831.00 | 19967.00 | Uruguay | Republic
3492 | UY | North America | North America | 9363520.00 | 1776 | 278357000 | 77.1 | 8510700.00 | 8110900.00 | United States | Federal Republic
3813 | UZ | Asia | Southern and Central Asia | 447400.00 | 1991 | 24318000 | 63.7 | 14194.00 | 21300.00 | Uzbekistan | Republic
3503 | UZ | Europe | Southern Europe | 0.40 | 1929 | 1000 | NULL | 9.00 | NULL | Santa Sede/Città del Vaticano | Independent Church State
3538 | VA | North America | Caribbean | 388.00 | 1979 | 114000 | 72.3 | 285.00 | NULL | Saint Vincent and the Grenadines | Constitutional Monarchy
3046 | VC | South America | South America | 912050.00 | 1811 | 24170000 | 73.1 | 95023.00 | 88434.00 | Venezuela | Federal Republic
3539 | VE | North America | Caribbean | 151.00 | NULL | 21000 | 75.4 | 612.00 | 573.00 | British Virgin Islands | Dependent Territory of the UK
3537 | VG | North America | Caribbean | 347.00 | NULL | 93000 | 78.1 | 0.00 | NULL | Virgin Islands of the United States | US Territory
4067 | VI | Asia | Southeast Asia | 331689.00 | 1945 | 79832000 | 69.3 | 21929.00 | 22834.00 | Việt Nam | Socialistic Republic
3770 | VN | Oceania | Melanesia | 12189.00 | 1980 | 190000 | 60.6 | 261.00 | 246.00 | Vanuatu | Republic
3537 | VU | Oceania | Polynesia | 200.00 | NULL | 15000 | NULL | 0.00 | NULL | Wallis-et-Futuna | Nonmetropolitan Territory of France
3536 | WF | Oceania | Polynesia | 2831.00 | 1962 | 180000 | 69.2 | 141.00 | 157.00 | Samoa | Parliamentary Monarchy
3169 | WS | Asia | Middle East | 527968.00 | 1918 | 18112000 | 59.8 | 6041.00 | 5729.00 | Al-Yaman | Republic
1780 | YE | Europe | Southern Europe | 102173.00 | 1918 | 10640000 | 72.4 | 17000.00 | NULL | Yugoslavija | Federal Republic
2722 | YU | Africa | Southern Africa | 1221037.00 | 1910 | 40377000 | 51.1 | 116729.00 | 129092.00 | South Africa | Republic
716 | ZA | Africa | Eastern Africa | 752618.00 | 1964 | 9169000 | 37.2 | 3377.00 | 3922.00 | Zambia | Republic
3162 | ZM | Africa | Eastern Africa | 390757.00 | 1980 | 11669000 | 37.8 | 5951.00 | 8670.00 | Zimbabwe | Republic
4068 | ZW
```

339 rows in set (0.002 sec)

MariaDB [(none)]>

2. Extrae la totalidad de los registros de la tabla de países para una inspección general:

```
SELECT * FROM world.country;
```

3. Utiliza la función de agregación para conocer el volumen total de registros:

```
SELECT COUNT(*) FROM world.country;
```

4. Inspecciona la estructura de las columnas para identificar los nombres técnicos de los campos:

```
SHOW COLUMNS FROM world.country;
```

239 rows in set (0.002 sec)

```

MariaDB [(none)]> SELECT COUNT(*) FROM world.country;
+-----+
| COUNT(*) |
+-----+
|      239 |
+-----+
1 row in set (0.002 sec)

MariaDB [(none)]> SHOW COLUMNS FROM world.country;
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| Code  | char(3) | NO | PRI | | |
| Name  | char(52) | NO | | | |
| Continent | enum('Asia','Europe','North America','Africa','Oceania','Antarctica','South America') | NO | | Asia | |
| Region | char(26) | NO | | | |
| SurfaceArea | decimal(10,2) | NO | | 0.00 | |
| IndepYear | smallint(4) | YES | | NULL | |
| Population | int(11) | NO | | 0 | |
| LifeExpectancy | decimal(3,1) | YES | | NULL | |
| GNP | decimal(10,2) | YES | | NULL | |
| GNPPold | decimal(10,2) | YES | | NULL | |
| LocalName | char(45) | NO | | | |
| GovernmentForm | char(45) | NO | | | |
| Capital | int(11) | YES | | NULL | |
| Code2 | char(2) | NO | | | |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
14 rows in set (0.001 sec)

MariaDB [(none)]>

```

5. Refina la consulta para extraer solo columnas específicas y utiliza un **Alias** para mejorar la presentación:

```
SELECT Name, Capital, Region, SurfaceArea AS "Surface Area", Population FROM world.country;
```

Saudi Arabia	Middle East	3173	2148636.00
Sudan	Northern Africa	3225	2505811.00
Senegal	Western Africa	3198	196722.00
Singapore	Southeast Asia	3208	611.00
South Georgia and the South Sandwich Islands	Antarctica	3203	3903.00
Saint Helena	Western Africa	3063	314.00
Switzerland and San Marino	Nordic Countries	3180	42422.00
Solomon Islands	Malaysia	3161	28596.00
Sierra Leone	Western Africa	3207	71740.00
El Salvador	Central America	645	21041.00
San Marino	Southern Europe	3171	61.00
Somalia	Eastern Africa	3214	637637.00
Saint Pierre and Miquelon	North America	3067	242.00
Sao Tome and Principe	Central Africa	3172	964.00
Suriname	South America	3243	163265.00
Slovakia	Eastern Europe	3209	49012.00
Slovenia	Southern Europe	3212	20256.00
Sweden	Nordic Countries	3048	449964.00
Swaziland	Southern Africa	3244	17364.00
Seychelles	Eastern Africa	3206	455.00
Syria	Middle East	3250	185130.00
Turks and Caicos Islands	Caribbean	3423	430.00
Chad	Central Africa	3137	1284500.00
Togo	Western Africa	3132	56735.00
Taiwan	Southeast Asia	3200	53115.00
Tajikistan	Southern and Central Asia	3261	143100.00
Tokelau	Polynesia	3133	12.00
Turkmenistan	Southern and Central Asia	3419	483100.00
East Timor	Southeast Asia	1522	14874.00
Tonga	Polynesia	3134	650.00
Trinidad and Tobago	Caribbean	3130	125000.00
Tunisia	Northern Africa	3349	163610.00
Turkey	Middle East	3155	774815.00
Tuvalu	Polynesia	3424	26.00
Taiwan	Eastern Asia	3263	36188.00
Tanzania	Eastern Africa	3306	883749.00
Uganda	Eastern Africa	3425	240330.00
Ukraine	Eastern Europe	3426	603700.00
United States Minor Outlying Islands	Micronesia/Caribbean	NULL	16.00
Uruguay	South America	3492	176016.00
United States	North America	3813	9363520.00
Uzbekistan	Southern and Central Asia	3503	447400.00
Wali Sze (Western City State)	Southern Europe	3581	20.40
Saint Vincent and the Grenadines	Caribbean	3066	385.00
Venezuela	South America	3539	912050.00
Virgin Islands, British	Caribbean	557	151.00
Virgin Islands, U.S.	Caribbean	4067	347.00
Vietnam	Southeast Asia	3770	331689.00
Vanuatu	Malaysia	3557	121250.00
Wallis and Futuna	Polynesia	3536	200.00
Samoa	Polynesia	3169	2831.00
Zanzibar	Middle East	1780	527960.00
Yugoslavia	Southern Europe	1792	102173.00
South Africa	Southern Africa	716	1221037.00
Zambia	Eastern Africa	3162	752610.00
Zimbabwe	Eastern Africa	4068	390757.00

6. Ordena los resultados de forma ascendente y descendente según la población:

- **Ascendente (por defecto):**

```
SELECT Name, Population FROM world.country ORDER BY Population;
```

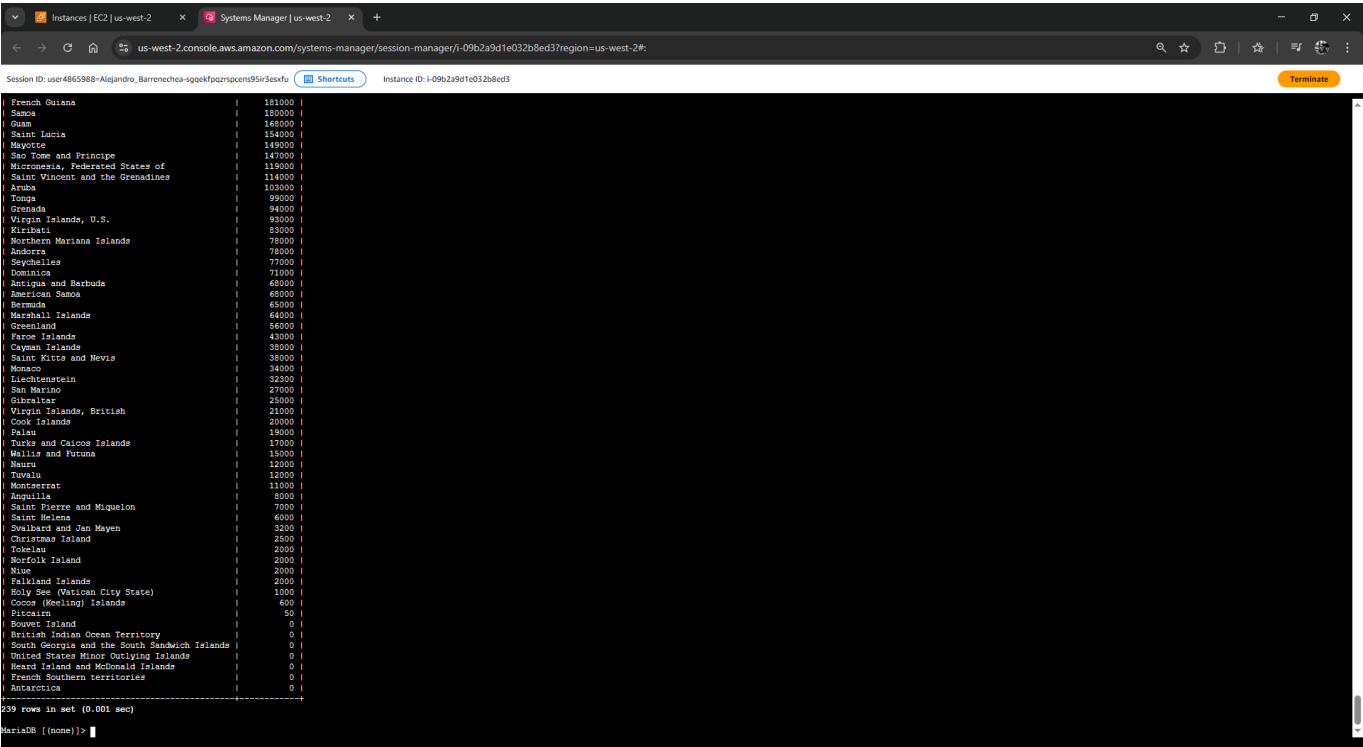
Syria	Middle East	16125000	
Kazakhstan	Central Asia	16023000	
Yemen	Western Asia	18112000	
Sri Lanka	Southeast Asia	18827000	
Australia	Oceania	18886000	
Mozambique	Southern Africa	19480000	
Ghana	Western Africa	20212000	
Saudi Arabia	Middle East	21607000	
Uganda	Eastern Africa	21778000	
Malaysia	Southeast Asia	22244000	
Taiwan	Eastern Asia	22256000	
Romania	Eastern Europe	22455000	
Afghanistan	South Asia	22720000	
Iraq	Middle East	23115000	
Nepal	South Asia	23930000	
North Korea	East Asia	24039000	
Venezuela	South America	24170000	
Uzbekistan	Central Asia	24318000	
Peru	South America	25662000	
Morocco	North Africa	28331000	
Sudan	North Africa	29490000	
Kenya	Eastern Africa	30080000	
Canada	North America	31147000	
Algeria	North Africa	31471000	
Tanzania	Eastern Africa	33517000	
Argentina	South America	37832000	
Poland	Central Europe	38653600	
Spain	South Europe	39441700	
South Africa	South Africa	40377000	
Colombia	South America	42321000	
Myanmar	South East Asia	45611000	
South Korea	East Asia	46844000	
Ukraine	Eastern Europe	50456000	
Congo, The Democratic Republic of the	Central Africa	51654000	
Italy	South Europe	57680000	
France	Western Europe	59225700	
United Kingdom	Western Europe	59623400	
Thailand	South East Asia	63399000	
Ethiopia	Eastern Africa	62865000	
Turkey	Western Asia	66591000	
Iran	Western Asia	67020000	
Egypt	North Africa	68470000	
Philippines	Southeast Asia	75947000	
Vietnam	South East Asia	79832000	
Germany	Central Europe	82164700	
Mexico	North America	98881000	
Nigeria	Western Africa	113506000	
Japan	East Asia	126714000	
Bangladesh	South Asia	129135000	
Russian Federation	North Asia	146334000	
Pakistan	South Asia	156483000	
Brazil	South America	170115000	
Indonesia	South East Asia	212107000	
United States	North America	278357000	
India	South Asia	1013662000	
China	East Asia	1277553000	

* ****Descendente (más poblados primero):****

```
```sql
```

```
SELECT Name, Population FROM world.country ORDER BY Population DESC;
```

```
```
```



Uso de Filtrado Avanzado (Cláusula WHERE)

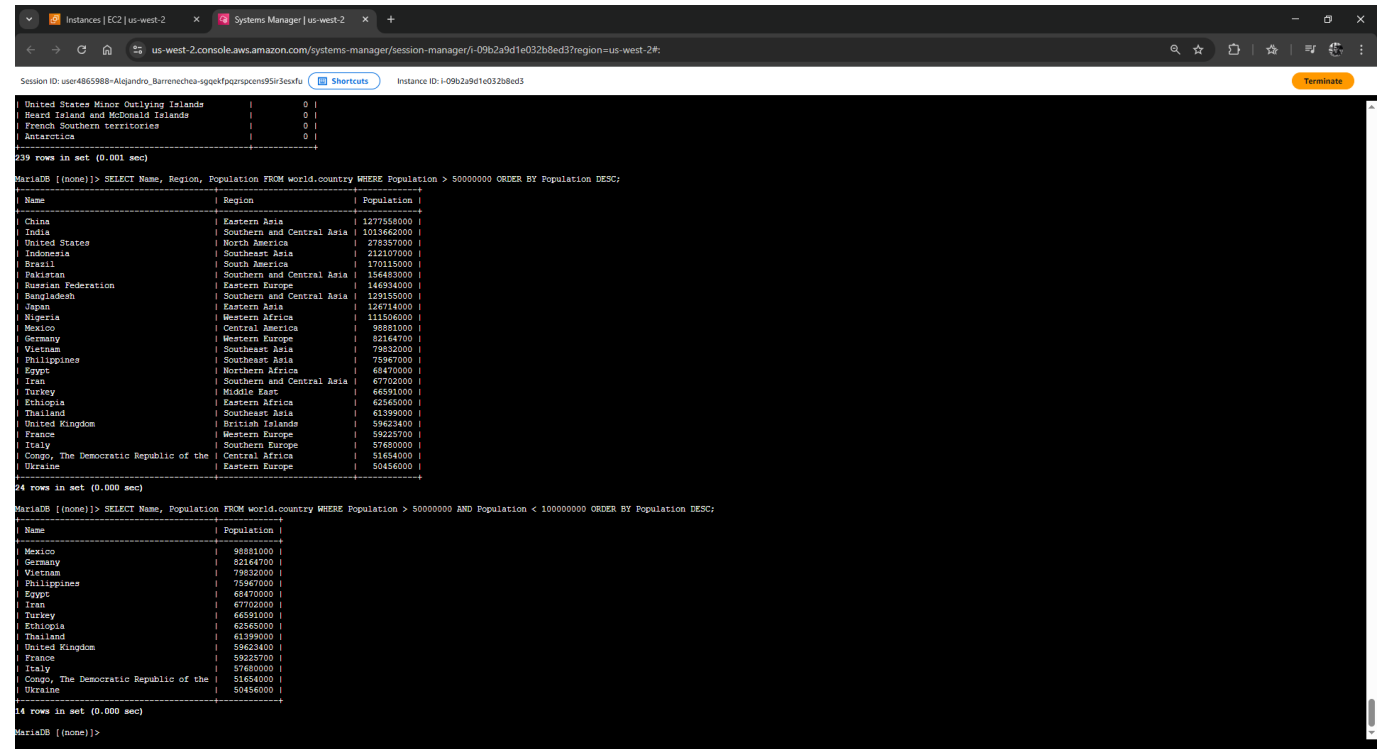
Para responder a criterios específicos de negocio, aplica filtros lógicos:

- 1. Filtra países con población superior a 50 millones, ordenados del más grande al más pequeño:

```
SELECT Name, Region, Population FROM world.country WHERE Population > 50000000 ORDER BY Population DESC;
```

- 2. Aplica un rango de población utilizando el operador lógico AND:

```
SELECT Name, Population FROM world.country WHERE Population > 50000000 AND Population < 100000000 ORDER BY Population DESC;
```



Challenge (Desafío Técnico)

Pregunta: ¿Qué país en "Southern Europe" (Europa del Sur) tiene una población mayor a 50,000,000?

Procedimiento: Debes combinar un filtro de cadena de texto (string) con un filtro numérico.

- 1. Escribe y ejecuta la siguiente consulta:

```
SELECT Name, Region, Population
FROM world.country
WHERE Region = 'Southern Europe' AND Population > 50000000;
```

Analiza el resultado: Deberías obtener a **Italy** (Italia) como el único registro que cumple ambos criterios.

```
Instances | EC2 | us-west-2 x Systems Manager | us-west-2 x +
us-west-2.console.aws.amazon.com/systems-manager/session-manager/i-09b2a9d1e032b8ed3?region=us-west-2#
Session ID: user4865988-Alejandro_Barrenechea-qqekfpqzrscen95ir3csfu Shortcuts Instance ID: i-09b2a9d1e032b8ed3 Terminate

China	Eastern Asia	1277558000
India	Southern and Central Asia	1013662000
United States	North America	278357000
Indonesia	Southeast Asia	212107000
Brazil	South America	170115000
Pakistan	Southern and Central Asia	156483000
Russian Federation	Eastern Europe	146934000
Bangladesh	Southern and Central Asia	129155000
Japan	Eastern Asia	126714000
Nigeria	Western Africa	115506000
Mexico	Central America	98881000
Germany	Western Europe	82164700
Vietnam	Southeast Asia	79832000
Philippines	Southeast Asia	75967000
Egypt	Northern Africa	68470000
Iran	Southern and Central Asia	67702000
Turkey	Middle East	66591000
Ethiopia	Eastern Africa	62565000
Thailand	Southeast Asia	61399000
United Kingdom	British Islands	59623400
France	Western Europe	59225700
Italy	Southern Europe	57689000
Congo, The Democratic Republic of the	Central Africa	51654000
Ukraine	Eastern Europe	50456000

24 rows in set (0.000 sec)

MariaDB [(none)]> SELECT Name, Population FROM world.country WHERE Population > 50000000 AND Population < 100000000 ORDER BY Population DESC;
+-----+-----+
| Name | Population |
+-----+-----+
Mexico	98881000
Germany	82164700
Vietnam	79832000
Philippines	75967000
Egypt	68470000
Iran	67702000
Turkey	66591000
Ethiopia	62565000
Thailand	61399000
United Kingdom	59623400
France	59225700
Italy	57689000
Congo, The Democratic Republic of the	51654000
Ukraine	50456000
+-----+-----+
14 rows in set (0.000 sec)

MariaDB [(none)]> SELECT Name, Region, Population
-> FROM world.country
-> WHERE Region = 'Southern Europe' AND Population > 50000000;
+-----+-----+-----+
| Name | Region | Population |
+-----+-----+-----+
| Italy | Southern Europe | 57689000 |
+-----+-----+-----+
1 row in set (0.000 sec)

MariaDB [(none)]>
```

Respuestas Analíticas y Conclusiones

- **¿Cuál es la diferencia entre COUNT(*) y COUNT(Columna)?** COUNT(*) cuenta todas las filas del conjunto de resultados, incluyendo aquellas que puedan tener valores nulos en algunas celdas. COUNT(Population), por otro lado, solo contará los registros donde la columna "Population" no sea NULL.
- **Importancia de los Alias (AS):** En el desarrollo de aplicaciones y reportes, los nombres de las columnas en la base de datos suelen ser técnicos o abreviados (ej. SrfArea). El uso de AS permite que el resultado entregado al usuario final o a la interfaz sea comprensible y profesional sin cambiar la estructura real de la tabla.
- **Eficiencia del filtrado con WHERE:** Como ingeniero de AWS, debes saber que filtrar los datos en el motor de base de datos (lado del servidor) es mucho más eficiente que traer todos los datos a la aplicación y filtrarlos allí. Esto reduce la latencia de red y el consumo de memoria del cliente.

¡Felicidades! Has completado con éxito las operaciones de consulta y análisis de datos en un entorno MySQL gestionado.